Impact: The Skyborg–Valkyrie combination validated the concept of attritable UAVs, low-cost, semi-expendable combat drones capable of performing high-risk missions, ushering in a new era of distributed airpower and network-centric warfare.

5.8.2 Case study 2: Humanitarian response:

Swarm drones in post-disaster recovery (2023)

In 2023, following a devastating earthquake in Turkey and Syria, an international coalition deployed a swarm of AI-coordinated UAVs based on fourth-generation architecture. The system integrated autonomous navigation, multi-sensor mapping, and real-time data fusion to survey affected areas, detect survivors through thermal imaging, and deliver medical supplies. The drones operated in three tiers: high-altitude reconnaissance, mid-level terrain mapping, and low-altitude logistics delivery coordinated through a cloud-based mission AI that distributed tasks dynamically.

Impact: The swarm achieved rapid situational awareness, mapping 250 square kilometers in under 24 hours and identifying over 700 survivor locations. The mission demonstrated the real-world humanitarian potential of autonomous UAV swarms, leading to global policy discussions on AI use in humanitarian logistics and autonomous disaster response frameworks.

5.8.3 Case study 3: Strategic surveillance: FCAS and remote carrier integration (2021–present)

Within Europe, the Future Combat Air System (FCAS) initiative showcased the power of manned–unmanned integration through Remote Carrier (RC) drones operating under AI-supervised control. In simulated joint operations, RC UAVs supported manned Next Generation Fighters (NGF) by providing